NITHISH KUMAR VEMU

Arlington, TX | vemunithishkumar@gmail.com| LinkedIn | +1 (660) 528-8942

PROFESSIONAL SUMMARY

PLM Change Analyst and Enterprise Systems professional with **3+ years of experience** supporting **Product Lifecycle Management (PLM)** and **Engineering Change Management** in manufacturing, aerospace, and enterprise environments. Strong hands-on experience with **ECO, ECR, ECN, MCO execution**, **BOM management (EBOM, MBOM, SBOM)**, **revision control**, and **configuration management**. Experienced in working with **Arena-style cloud PLM workflows**, cross-functional engineering teams, and manufacturing partners. Background in **Mechanical Engineering** with a Master's in **Information Systems**, passionate about clean energy, process optimization, and scalable product operations.

TECHNICAL SKILLS

- **PLM Tools**: Arena PLM (workflow-based experience), PLM/PDM Systems
- **Change Management:** ECO, ECR, ECN, MCO, Deviations, Change Control Boards (CCB)
- **BOM Management:** EBOM, MBOM, SBOM, Part Numbering, Revision Control
- **Configuration & Release:** Part Lifecycle States, Configuration Management, Product Release
- **Manufacturing & NPI:** New Product Introduction (NPI), High-Volume Manufacturing, Contract Manufacturers
- **Cross-Functional Collaboration**: Engineering, Manufacturing, Quality, Supply Chain, Operations
- **Documentation & Training**: SOPs, PLM Policies, User Training, Best Practices
- **Data & Systems**: Engineering Data Management, Structured Data, SQL, Master Data Governance
- **CAD Exposure:** Creo, 3D Modeling, 2D Drafting, Assembly Models
- **Quality & Compliance:** Traceability, Audit Support, NDT Awareness
- **Methodologies:** Agile, Waterfall, Continuous Improvement

PROFESSIONAL EXPERIENCE

Benda Infotech — Jul 2024 -Present

Enterprise Systems Analyst — Remote (USA)

- Supported and enhanced **enterprise systems handling structured lifecycle data**, comparable to engineering part records, metadata, and revision histories.
- Executed **change requests, impact analysis, validation, and controlled releases**, aligning closely with **PLM change control processes.**
- Collaborated **with cross-functional teams (engineering, QA, operations) to review system changes**, approvals, and deployments.
- Performed unit and integration testing to validate system stability following enhancements and fixes.
- Created and maintained technical documentation, SOPs, and user guides, supporting onboarding and operational consistency.
- Assisted with production issue triage, root cause analysis, and resolution to improve system reliability.

BAIF Research Development Foundation — Aug 2022 – Jul 2023

Project Manager — Hyderabad, India

- Led enterprise platform development supporting structured data workflows, traceability, and lifecycle governance.
- Managed change requests, release coordination, and documentation, ensuring compliance with defined processes.
- Acted as a liaison between engineering, IT, and business stakeholders, similar to PLM analyst responsibilities.
- Improved system performance and usability through process optimization and continuous improvement initiatives.
- Oversaw change requests, release coordination, and technical documentation, ensuring controlled updates, compliance with internal processes, and clear alignment among all stakeholders.

MSME-TOOL ROOM — Mar 2022 – Jun 2022

Product Developer/ Internship — Hyderabad, India

- Performed iterative design changes, enabled reverse engineering, and successfully scanned existing prototypes using **3D scanning and 3D printing technologies** to develop functional products.
- Participated in **reverse engineering workflows** to convert physical components into manufacturable digital models.
- Learned and applied **non-destructive testing (NDT) techniques** in practical environments, ensuring product safety, structural integrity, and cost-effective quality control throughout the development lifecycle.
- Assisted in evaluating **material behavior and tolerances** during testing and validation stages.
- Enhanced manufacturing accuracy and expedited product development from concept to production-ready designs by participating in

high-precision 3D printing workshops using DLP technology.
- Collaborated with engineers and technicians to support **prototype refinement, documentation, and production readiness**.

TIRVEN INDUSTRIES PVT LTD **Jun 2021 – Nov 2021**
Fabrication Technician Hyderabad, India
- Worked as an active team member and **led a team of 5 members** on an **aircraft project related to defense research**.
- Supported **engineering design analysis and optimization** of an aircraft rudder by reducing weight and enhancing performance using **carbon fabric materials**.
- Assisted in evaluating **material selection and fabrication methods** to meet performance and safety requirements.
- Fabricated **alloy steel components** and supported precise **adhesive bonding processes** to guarantee structural integrity and performance in critical applications.
- Followed **quality, safety, and manufacturing standards** during fabrication and assembly activities.
- Coordinated with engineering and production teams to ensure smooth workflow and timely task completion.

EDUCATION

Northwest Missouri State University **May 2025**
M.S. in Information Systems — GPA: 3.87 / 4.00 Maryville, MO

Vignan Institute of Technology and Science **Sep 2022**
B.E. in Mechanical Engineering — GPA: 3.00 / 4.00 Telangana, India

CERTIFICATIONS

- **Product Development Using 3D Scanning and 3D Printing** — MSME-TOOL ROOM, 2022
- **Oracle Database SQL Certified Expert** —HACKERRANK, 2024
- **NDT In Quality & Energy Sector —** VIDAL NDT, 2022
- **Fabrication Of Tail Cone Of Aircraft —** TIRVEN INDUSTRIES PVT.LTD, 2021

ACADEMIC & RESEARCH PROJECTS

- **Design, Manufacturing and Analysis of Waveguide for Parabolic Antenna Using EDM Process** **Mar 2022 - Jun 2022**

  Focuses on the design and manufacturing of a waveguide for parabolic antenna applications to ensure efficient electromagnetic wave transmission with minimal signal loss. The project involved iterative design modifications and cross-section optimization using CAD tools, followed by precision manufacturing through milling and EDM. NDT techniques were applied to validate structural integrity, surface accuracy, and quality throughout the manufacturing process. High-precision machining enabled smooth internal surfaces, ensuring disturbance-free electromagnetic wave propagation, improved performance, and reliable operation in wireless communication systems.
- **Design and Additive Manufacturing of Drone Wing** **Oct 2021 – Feb 2022**

  Designed and developed a lightweight drone wing using additive manufacturing to achieve an optimal strength-to-weight ratio and aerodynamic performance. Utilized 3D printing with internal lattice structures to reduce material usage while maintaining structural integrity and stable flight performance.
- **Design, Fabrication and Optimization of Aircraft Rudder** **Jul 2021 - Sep 2021**

  Focuses on enhancing aerodynamic efficiency and structural integrity of an aircraft rudder through design analysis and material optimization. The project involved studying weight reduction techniques by incorporating lightweight carbon fabric materials while maintaining strength and performance requirements. Fabrication processes included precise bonding and assembly of alloy steel and composite components to ensure structural stability under operational loads. Emphasis was placed on improving control stability, durability, and fuel efficiency, resulting in a highly efficient and reliable rudder suitable for modern aircraft applications.